THE GALOIS GROUP WITH ITS ACTION TOPOLOGY IS A PROFINITE TOPOLOGICAL GROUP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The argument is nonstandard by design: the topology is defined before the inverse limit presentation, so profinite-ness is proved by exhibiting Gal(E/F) as a closed subset of the explicit profinite limit G = lim G(S).

Editorial note. The label lemma-galois-profinite names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Topology, Example topology-example-automorphisms-of-a-set: the compact open topology on Map(E, E) is the universal topology making the action Map(E, E) x E -> E continuous (E discrete), and Aut(E) with the compact open topology is a topological group; cited as standard facts of the sibling Topology chapter
- Topology, Lemma topology-lemma-profinite-group: a topological group whose underlying space is profinite is a profinite group
- Same chapter: lemma-algebraic-extension-self-map (an F-algebra self-map of an algebraic extension is an automorphism)

Source reference. Original reference 0BMD: sources/stacks-fields/pinned-original-source.tex, source lines 1024–1024. Prerequisite pointer invoked at 3159; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0BMC: sources/stacks-chow/tags.txt, source lines 14625–14625. Prerequisite pointer invoked at 3097; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BR1: sources/stacks-chow/tags.txt, source lines 14754–14754. Prerequisite pointer invoked at 3122; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0B1Y: sources/stacks-chow/tags.txt, source lines 13911–13911. Prerequisite pointer invoked at 3113; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let E/F be a Galois extension. Endow $Gal(E/F)$ with the coarsest topology such that*

$$Gal(E/F) \times E \longrightarrow E$$

is continuous when E is given the discrete topology. Then

1. *for any topological space X and map $X \to Gal(E/F)$ such that the action $X \times E \to E$ is continuous the induced map $X \to Gal(E/F)$ is continuous,*
2. *this topology turns $Gal(E/F)$ into a profinite topological group.*

Proof. Throughout this proof we think of E as a discrete topological space. Recall that the compact open topology on the set of self maps $\mathrm{Map}(E, E)$ is the universal topology such that the action $\mathrm{Map}(E, E) \times E \to E$ is continuous. See Topology, Example 0BMC for a precise statement. The topology of the lemma on $\mathrm{Gal}(E/F)$ is the induced topology coming from the injective map $\mathrm{Gal}(E/F) \to \mathrm{Map}(E, E)$. Hence the universal property (1) follows from the corresponding universal property of the compact open topology. Since the set of invertible self maps $\mathrm{Aut}(E)$ endowed with the compact open topology forms a topological group, see Topology, Example 0BMC, and since $\mathrm{Gal}(E/F) = \mathrm{Aut}(E/F) \to \mathrm{Map}(E, E)$ factors through $\mathrm{Aut}(E)$ we obtain a topological group. In other words, we are using the injection

$$\mathrm{Gal}(E/F) \subset \mathrm{Aut}(E)$$

to endow $\mathrm{Gal}(E/F)$ with the induced structure of a topological group (see Topology, Section 0B1Y) and by construction this is the coarsest structure of a topological group such that the action $\mathrm{Gal}(E/F) \times E \to E$ is continuous.

To show that $\mathrm{Gal}(E/F)$ is profinite we argue as follows (our argument is necessarily nonstandard because we have defined the topology before showing that the Galois group is an inverse limit of finite groups). By Topology, Lemma 0BR1 it suffices to show that the underlying topological space of $\mathrm{Gal}(E/F)$ is profinite. For any subset $S \subset E$ consider the set

$$G(S) = \{f : S \to E \mid \begin{matrix} f(\alpha) \text{ is a root of the minimal polynomial} \\ \text{of } \alpha \text{ over } F \text{ for all } \alpha \in S \end{matrix}\}$$

Since a polynomial has only a finite number of roots we see that $G(S)$ is finite for all $S \subset E$ finite. If $S \subset S'$ then restriction gives a map $G(S') \to G(S)$. Also, observe that if $\alpha \in S \cap F$ and $f \in G(S)$, then $f(\alpha) = \alpha$ because the minimal polynomial is linear in this case. Consider the profinite topological space

$$G = \lim\nolimits_{S \subset E \text{ finite}} G(S)$$

Consider the canonical map

$$c : \mathrm{Gal}(E/F) \longrightarrow G, \quad \sigma \longmapsto (\sigma|_S : S \to E)_S$$

This is injective and unwinding the definitions the reader sees the topology on $\mathrm{Gal}(E/F)$ as defined above is the induced topology from G. An element $(f_S) \in G$ is in the image of c exactly if (A) $f_S(\alpha) + f_S(\beta) = f_S(\alpha + \beta)$ and (M) $f_S(\alpha) f_S(\beta) = f_S(\alpha\beta)$ whenever this makes sense (i.e., $\alpha, \beta, \alpha + \beta, \alpha\beta \in S$). Namely, this means $\lim f_S : E \to E$ will be an F-algebra map and hence an automorphism by Lemma 0BMD. The conditions (A) and (M) for a given triple (S, α, β) define a closed subset of G and hence $\mathrm{Gal}(E/F)$ is homeomorphic to a closed subset of a profinite space and therefore profinite itself. □